



مدارس الرواد العالمية
AL-ROWAD INTERNATIONAL SCHOOLS

CYBERSECURITY FOUNDATIONS

Student's Guide to Cybersecurity



Higher Grades

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This book has been developed as an educational resource to help learners build a comprehensive understanding of cybersecurity concepts, principles, and best practices. Every effort has been made to ensure that the information presented is accurate and up to date at the time of publication. However, as cybersecurity is a rapidly evolving field, technologies, standards, and best practices may change over time.

The practical activities, examples, and case studies included in this book are intended for educational purposes and should be performed only in authorized learning environments. Readers are encouraged to apply the knowledge gained from this book responsibly, ethically, and in accordance with applicable laws, institutional policies, and professional standards. The author and publisher are not responsible for any misuse of the information contained in this publication.

Readers are encouraged to continue exploring current cybersecurity developments through further study and trusted professional resources. Feedback and suggestions for improving future editions of this book are always welcome. It is our hope that this book serves as a valuable learning companion and inspires a lifelong commitment to ethical cybersecurity practices.



PREFACE



Cybersecurity is one of the fastest-growing and most important fields in today's digital world. This book has been carefully designed to take learners on a structured journey from the fundamentals of cybersecurity to advanced security concepts, providing a strong foundation for academic study and professional growth.

A unique feature of this book is its visual learning approach. Every theoretical concept has been transformed into a visualization and reinforced with step-by-step examples and real-world cybersecurity scenarios, making complex cybersecurity concepts easier to understand, improving knowledge retention, and helping learners connect theory with practical, real-world applications.

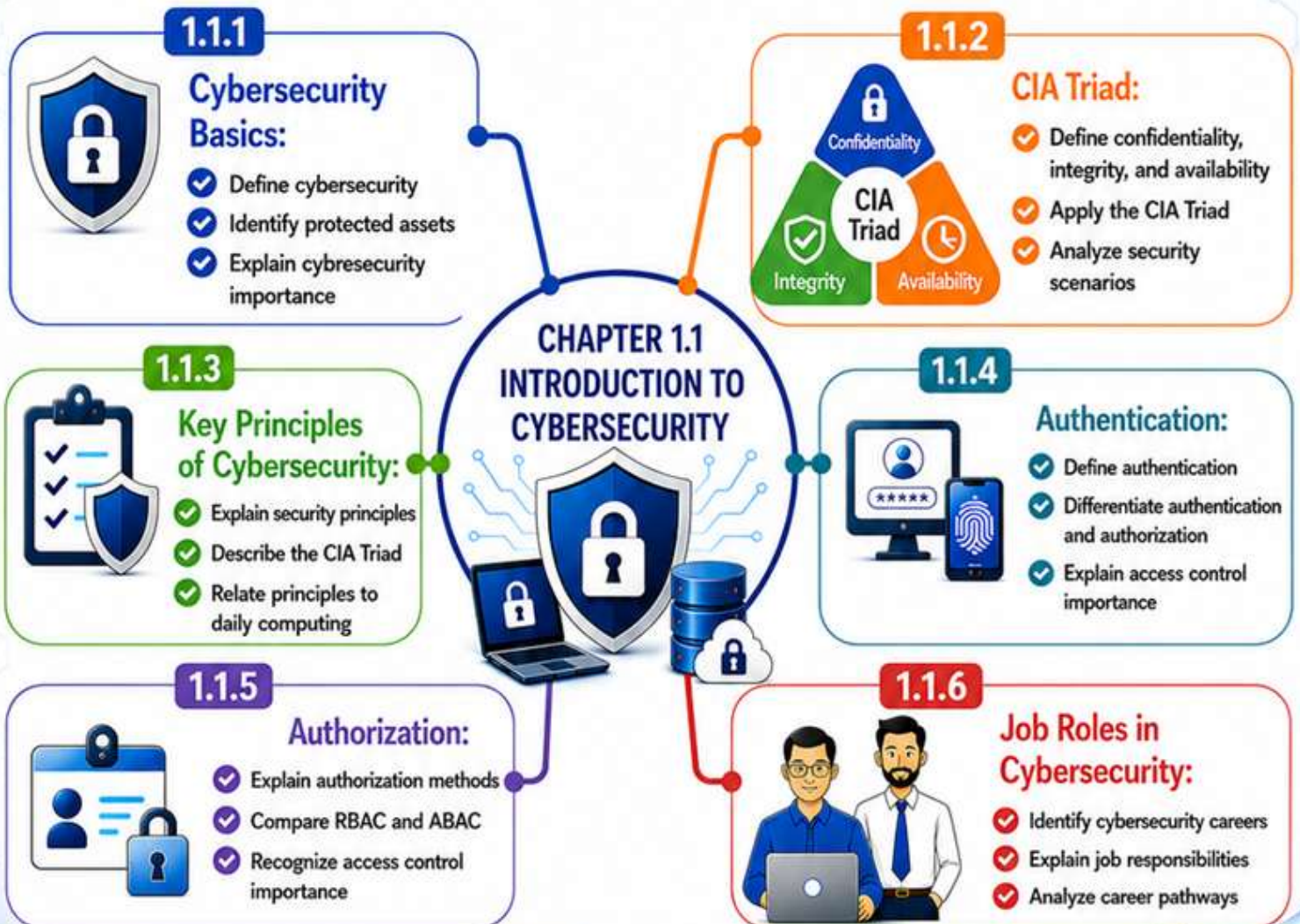
This book is dedicated to every student who is passionate about learning, exploring, and building a future in cybersecurity. Whether you are beginning your journey or expanding your knowledge, may this book inspire confidence, curiosity, ethical responsibility, and a commitment to lifelong learning as you prepare to become the next generation of cybersecurity professionals.

Lone Firdous
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UNIT 1: FUNDAMENTALS OF CYBERSECURITY

CHAPTER 1: INTRODUCTION TO CYBERSECURITY

Chapter 1 provides the basic concepts of Cybersecurity and how individuals and organizations protect computer, networks, and digital information from cyber threats. This course will cover the Overview of Cybersecurity, Key Concepts of Cybersecurity, Confidentiality, Integrity and Availability (C.I.A) Triad, which is used as the key foundation for information security, Authentication, Authorization, Role-Based Access Control (RBAC) and Attribute-Based Access Control (ABAC) and how these permission mechanisms help us gain secure access to the digital assets. Lastly, you will learn about the various Job Roles in Cybersecurity and how cybersecurity professionals collaborate to secure modern information systems & the networks.





❖ 1.1.1 CYBERSECURITY BASICS

Cybersecurity involves the protection of computers, networks, software and information from harm, theft or unauthorized access. It includes the **technologies**, policies and procedures used to protect digital systems from cyber threats. One of the primary functions of cybersecurity is to maintain the **confidentiality**, integrity and availability of information.

► **Example:** On a typical morning, Ahmed turns on his laptop before starting his schoolwork. His antivirus software automatically scans the computer and quickly blocks a malicious file that could have infected the system. Later, his mother uses the family's banking app to transfer money. Behind the scenes, encryption protects her personal information while the bank's firewall blocks unauthorized access from hackers. At school, students connect to the school's Wi-Fi to complete online assignments, but the school's cybersecurity system automatically blocks harmful websites and protects student records from unauthorized users. Although none of them notice these security measures, cybersecurity works continuously in the background to keep their devices, information, and online activities safe.

1 Antivirus protects the computer.

Antivirus software scans the computer and blocks malware.



2 The bank secures online transactions.

Encryption and firewalls protect personal information from hackers.



3 School cybersecurity keeps students safe.

The school network blocks harmful websites and protects student records.



4 Cybersecurity works in the background every day.

It keeps devices, information, and online activities safe without us even noticing.



■ Protecting Digital Information

The digital **assets** consist of files and documents, emails, passwords, photos and videos, databases – all the data that you keep in electronic form. Organizations gather and keep large amounts of private **data**, financial information, and health records. This is where

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cybersecurity comes in, to help protect this information using encryption, access control and backup **systems** as some measures.



■ Protecting Systems and Networks

A **network** is collection of devices that are physically connected together such that they can easily communicate and share information. Cybersecurity prevents **systems** from malware, computer viruses, spyware, worms and hackers. Security **technologies**, including firewalls, Intrusion Detection Systems (IDS), and antivirus software can detect and prevent **attacks** before causing damage. Together, these security measures help maintain the confidentiality, integrity, and availability of network resources.

■ Preventing and Responding to Cyberattacks

The **preventive** measures, plus the **response** level, is cybersecurity. Security **monitoring**, vulnerability assessment and patch management are put in place to reduce what an **attack** can do with the least effort possible on organizations. However, no **system** is foolproof, which is why organizations also build **incident** response & disaster recovery plans and have backup **data**. A well-prepared response enables organizations to recover quickly and continue operations with minimal downtime. Regular testing and updating of security procedures ensure that organizations remain prepared to handle emerging cyber threats effectively.

■ Why do we need Cybersecurity in daily life?

In fact, everything we do today is either enabled or dependent on **technology** — from online banking to shopping to social media and more; simply said there is hardly anything in sight which does not use any kind of technological **device**. If not, **malicious** people might get access to our passwords, bank details and personal information. Additionally, **businesses** rely on cybersecurity to keep their **customers** secure too, and this is how they ensure that they gain **trust**. As technology has continued to **evolve** there is increasing **exposure** to it for individuals, corporations and governments. Practicing good cybersecurity habits helps create a safer and more secure digital environment for everyone.



❖ 1.1.2 CIA Triad

One of the fundamental **models** in cybersecurity is the CIA Triad (Confidentiality, Integrity and Availability). It tells about the top three **objectives** of information security, which are Confidentiality, Integrity and Availability. These are the three **principles** every security system is designed to protect.

► **Example:** A hospital encrypts patient records so that only doctors and authorized staff can read them, maintaining confidentiality. It uses access controls and audit logs to prevent unauthorized changes to medical records, protecting integrity. The hospital also keeps backup servers and emergency power systems to ensure patient information is available even during a system failure.



■ Confidentiality

Confidentiality protects **information** so that it is only available to proper users. It is important to keep **sensitive** information confidential so that unauthorized users do not have access to your social security number, credit card numbers, bank accounts or **health-related** records. Enforce **control** on sensitive data: Organizations implement encryption, password protection, access control and Multi-Factor Authentication (MFA) to implement confidentiality. Data **breaches**, identity theft and loss of customer confidence are examples of privacy violations.

■ Integrity

Integrity is to ensure that the **information** remains complete and correct unless updated by an authorized person. Users should be **confident** that the information they access is accurate and has not been altered. These **technologies** and processes -- hashing, digital signatures and audit logs -- ensure that data integrity is preserved. When **integrity** is lost, inadequate and incorrect information may lead to operational problems.

■ Availability

Availability is to be able to access the **systems** and information whenever an authorized user needs them. To ensure **availability**, organizations deploy data backups, **redundant** servers, cloud computing and disaster recovery plans.

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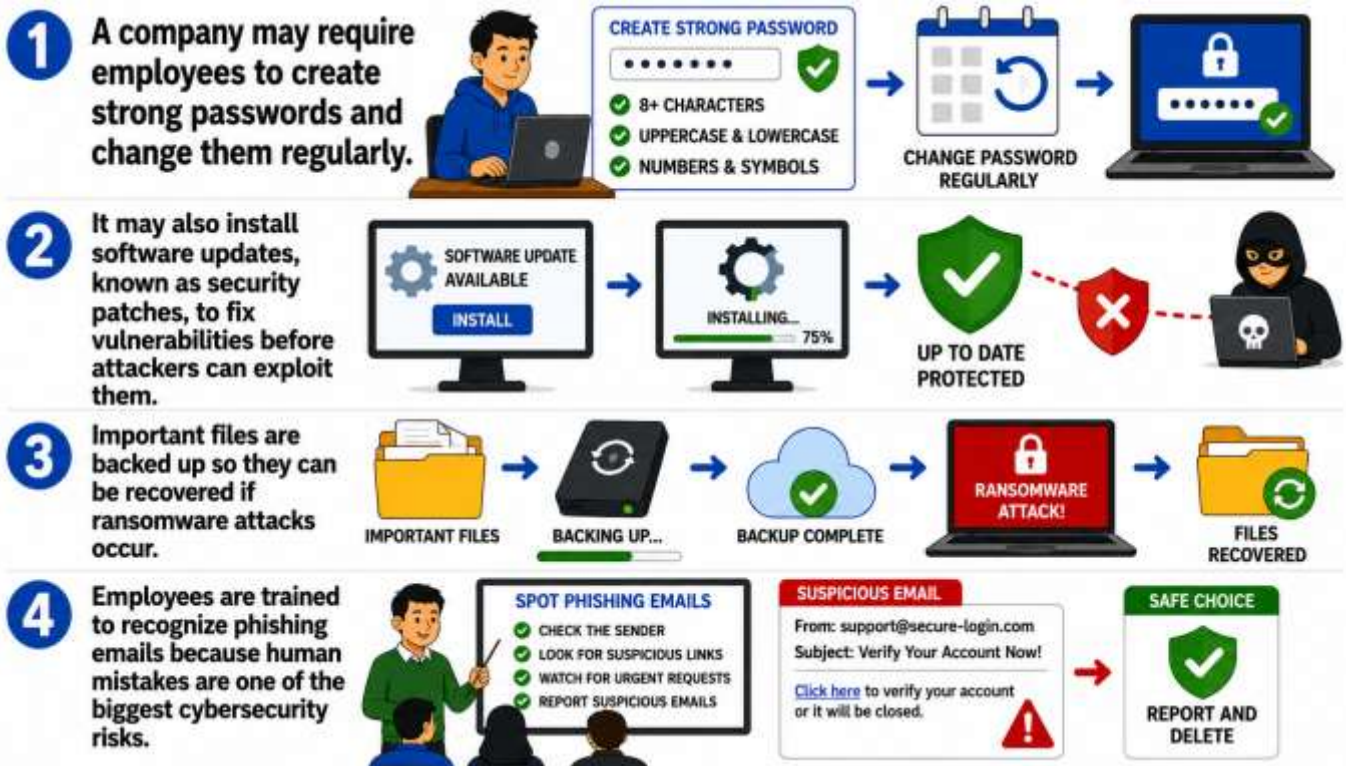


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Availability can be reduced as a result of **hardware** failures, natural disasters and Denial-of-Service (DoS) attacks. Availability is **crucial** since businesses and services rely on constant access to data.

❖ 1.1.3 KEY PRINCIPLES OF CYBERSECURITY

Cybersecurity is based on a standard set of best practices that are the fundamental principles required to protect your information systems against cyber attacks. **Principles** highlight risk minimization, need to know access and conduct operations safely. Examine the **illustration** below to see how cybersecurity principles are applied in practice.



Identify and Reduce Risks

Risk management is the process of identifying, analyzing and minimizing such **security** threats which may affect an organization. Risk **assessments** will additionally gauge the probability of a **threat** occurring as well as how detrimental it would be if that threat **occurs**. To combat such **threats**, organizations implement firewalls and access restrictions. Organizations with strong risk **management** are more equipped to confront cyber threats.

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■ Use Multiple Security Layers

Defense in Depth is a security **approach** where multiple layers of **defense** are used rather than relying on one single defense control. The different **layers** can include firewalls, antivirus, encryption, network monitoring and authentication systems. In case of multiple security layers, if one layer fails to achieve the task, **another** will do it. Such a **method** complicates things for the attackers because it becomes more difficult to access important systems and data.

■ Principle of Least Privilege

According to the Principle of Least **Privilege** (PoLP), users must only be given **permissions** needed to do their jobs. No unnecessary **access** to systems and confidential information should be given to employees. Restricting **permissions** also greatly reduces the risk of insider threats, unauthorized access and human **errors**. It is utilized in access **control** systems and network administration.

■ Security Awareness

Security awareness is the **knowledge** to educate users about cyber threats and safe computing practices. **Training** employees to identify phishing emails, social engineering attacks and suspicious websites is important. Through security awareness **programs**, users are also trained to create secure **passcodes** and safeguard sensitive data. Regular **training** is important because human **errors** are one of the main causes of cyber incidents.

■ Keep Systems Updated

Software **developers** constantly release security patches to repair **vulnerabilities** discovered in their programs. Failure to install these **updates** would allow attackers to use those **weaknesses**. One of the oldest yet most effective cybersecurity **practices** is keeping the software up to date. Routine **patch** management substantially decreases the likelihood of cyberattacks.

SESSION REVIEW



1.1.1 CYBERSECURITY BASICS

Cybersecurity Basics



- Protects computers, networks and data from cyber threats and unauthorized access.

Protecting Digital Information



- Protects files, emails, passwords and personal data from unauthorized access.

Protecting Systems and Networks



- Prevents malware, viruses, spyware and hacker attacks.

Preventing and Responding to Cyberattacks



- Reduces attacks through monitoring, vulnerability assessment and patching.

Why Cybersecurity is Important



- Protects personal, financial and business information from cybercrime.

1.1.2 CIA TRIAD

CIA Triad

- Core cybersecurity model based on Confidentiality, Integrity and Availability.
- Guides the protection of information systems and digital assets.



Confidentiality

- Ensures only authorized users can access sensitive information. Uses encryption, passwords, access control and MFA.



Integrity

- Ensures data remains accurate, complete and unaltered. Uses hashing, digital signatures and audit logs.



Availability

- Ensures systems and data are accessible when needed. Uses backups, redundant servers and disaster recovery plans.

1.1.3 KEY PRINCIPLES OF CYBERSECURITY

Risk Management



- Identifies and evaluates security risks before they become threats.

Defense in Depth



- Uses multiple security layers instead of relying on one defense.

Principle of Least Privilege (PoLP)



- Users receive only the permissions required for their jobs.

Security Awareness



- Trains users to recognize phishing, scams and social engineering attacks.

Regular Updates and Patching



- Installs security updates to fix software vulnerabilities.